Week 2: Literature Review

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I. Project Title

CVaR-MPPI for Highway Lane-Change Planning.

II. Main Idea

I propose a risk-sensitive highway lane-change planner that extends the Model Predictive Path Integral (MPPI) control framework with a Conditional Value-at-Risk (CVaR) objective. The goal is to replace risk-neutral rollout scoring with a sampled tail-risk backup, so the planner avoids lane-change actions that are efficient on average but unsafe under rare traffic outcomes.

III. Literature Overview

This project connects sampling-based control, tail-risk optimization, and dynamic programming. Williams, Aldrich, and Theodorou derive MPPI as a real-time stochastic optimal control method that uses path-integral ideas, importance sampling, and parallel rollout computation [1]. Rockafellar and Uryasev define CVaR as the conditional expectation of losses beyond VaR and show how CVaR can be optimized using an auxiliary function that supports scenario-based computation [2]. Chow and Ghavamzadeh extend CVaR optimization to MDPs through a mean-CVaR formulation and propose policy-gradient and actor-critic algorithms for risk-sensitive policies [3]. Chow, Tamar, Mannor, and Pavone provide the most direct dynamic programming foundation by formulating CVaR MDPs, deriving a CVaR Bellman operator, and presenting approximate value iteration with error guarantees [4]. Yin, Zhang, and Tsiotras provide the implementation bridge by adding Monte Carlo CVaR evaluation to MPPI and using the resulting risk value to filter sampled trajectories [5]. My project will use the CVaR Bellman view as the theoretical motivation and the RA-MPPI structure as the practical implementation path for highway lane-change planning.

IV. Highlighted Literature Review

A. Summary

Chow, Tamar, Mannor, and Pavone study risk-sensitive decision making in MDPs using a CVaR objective [4]. They show that CVaR can also be interpreted as expected cost under worst-case transition perturbations with a bounded error budget. They then derive a CVaR Bellman equation and an approximate value-iteration method with convergence and error guarantees.

B. Key Idea of the Work & Relevance to Your Project

The key idea is that CVaR can be used inside a sequential planning recursion, not only as an offline risk metric. In standard dynamic programming, a Bellman backup evaluates an action using expected future cost. In the CVaR MDP formulation of Chow, Tamar, Mannor, and Pavone, the backup evaluates the high-cost tail of future outcomes through an augmented state that includes the CVaR confidence level [4]. This is relevant to highway lane changes because the unsafe cases are rare but important. A nominal lane change may have low average cost, while a small set of traffic futures leads to a near collision. My project will not implement full tabular CVaR value iteration. Instead, it will approximate the same tail-risk principle inside MPPI by sampling stochastic traffic futures and using CVaR-based rollout costs. This preserves the dynamic programming motivation while keeping the implementation feasible for a two-month course project.

C. Main Limitations

The main limitation is that the highlighted work does not directly address continuous vehicle dynamics or real-time MPPI control. Its value-iteration method is designed for finite MDPs with an augmented confidence state, while highway lane-change planning requires fast online control in a continuous state space. I will address this gap by combining its CVaR Bellman motivation with the RA-MPPI implementation style of Yin, Zhang, and Tsiotras, where CVaR is estimated from disturbed rollouts and added to MPPI trajectory costs [5]. The final system will compare standard MPPI and sampled CVaR-MPPI in three-lane highway scenarios with uncertain surrounding vehicles.

References

[1] G. Williams, A. Aldrich, and E. A. Theodorou, "Model predictive path integral control: From theory to parallel computation," *Journal of Guidance, Control, and Dynamics*, vol. 40, no. 2, pp. 344–357, 2017.
[2] R. T. Rockafellar and S. Uryasev, "Optimization of conditional value-at-risk," *Journal of Risk*, vol. 2, no. 3, pp. 21–41, 2000.
[3] Y. Chow and M. Ghavamzadeh, "Algorithms for CVaR optimization in MDPs," in *Advances in Neural Information Processing Systems*, vol. 27, pp. 3509–3517, 2014.
[4] Y. Chow, A. Tamar, S. Mannor, and M. Pavone, "Risk-sensitive and robust decision-making: A CVaR optimization approach," in *Advances in Neural Information Processing Systems*, vol. 28, pp. 1522–1530, 2015.
[5] J. Yin, Z. Zhang, and P. Tsiotras, "Risk-aware model predictive path integral control using conditional value-at-risk," arXiv:2209.12842, 2022.

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